ASX Release

21 March 2024

GDM adds potential for Lithium, REE at its Cape Project

Great Divide Mining Ltd (the **Company** or **GDM**) (ASX:GDM), a new Queensland gold, antimony and critical metals explorer, provides an update following early exploration activities conducted at the Company's Cape Project within the New Goldfield tenement, EPM 26646.

Key Points:

- Lithium and Rare Earth Element (REE) potential has been identified in preliminary exploration results and desktop studies.
- Further exploration is required to assess the true potential for significant Lithium and REE mineralisation.
- Gold and base metals continue to be the primary focus of exploration at New Goldfield.
- Elsewhere, GDM continues to progress other projects including geology and mining studies towards assessing the commercial viability of Yellow Jack and planning for exploration at Coonambula and Devils Mountain pending field season commencing.

Commenting on this recent activity Chief Executive Officer of Great Divide Mining, Justin Haines, said:

> "Although we are in the infancy of our exploration at New Goldfield, early indications of prospectivity for Lithium and REE are encouraging and present a 'bonus' to GDM beyond the priority gold and base metal prospects on this tenement. We will continue to conduct on-site exploration at New Goldfield across our existing Gold and base metals targets as well as these new Lithium and REE opportunities."

On-Site Exploration at New Goldfield

The New Goldfield tenements (EPM 26646 and EPM Application 28913) are located within our Cape Project comprising 48 sub-blocks with an area of approximately 153 km^2 located ~230km northwest of Cairns in North Queensland. The New Goldfield area lies adjacent to the famous King Junction and Palmer River alluvial mining areas where significant coarse-grained alluvial gold was recovered in the mid-1800s.

GDM commissioned an assessment of New Goldfield for potential Lithium and REE mineralisation. A preliminary desktop study followed by a field trip was completed in late 2023.

Recent Lithium pegmatite discoveries in the nearby region indicates that North Queensland is fertile for Lithium-bearing LCT (Lithium-Cesium-Tantalum) pegmatite systems. Two known Lithium pegmatite projects occur to the south, within 300 km of the Cape Project: 1) Mount Surprise Lithium Project owned by Metalicity (ASX:MCT) located near Mount Surprise; and 2) Buchanan's Lithium Province owned by Strategic Metals Australia, located near Georgetown.

Field investigations at the Cape Project focused on inspecting the Chevy Creek Granite and its potential to host Lithium and REE. A number of preliminary targets were identified from earlier desktop studies then ground-checked in the field in late 2023. Several greisens and narrow pegmatites were located on the margins of the granite and rock chip sampled.

Using a Potassium/Rubidium (K/Rb) ratio as an indicator of the degree of fractionation and potential to host Lithium-bearing pegmatites, the preliminary rock chip assays are promising – with one sample resulting in a ratio of 75 (KJ008). That sample was taken within the greisen on the margins of the Chevy Creek Granite. Additionally, the presence of Beryl identified in at least one sample is consistent with the pegmatites being fertile and peraluminous. These encouraging results will be followed up in the future.

For personal use only

These new observations warrant further geological mapping, geochemical sampling and other exploration activities, to identify the true extent of the pegmatites and any associated Lithium/REE mineralisation.

Below is a table of sample K/Rb ratios:

SAMPLE	LOCATION GDA 2020			GPS	SAMPLE	RB_PPM	K_%	K/RB RATIO
	Zone	mE_Utm	mN_Utm	Elev_m	Type			
KJ 001	54	779887	8251433	150	Rock Chip	71.7	1.9	265
KJ 002	54	779999	8251472	152	Rock Chip	43.3	1.0	231
KJ 003	54	782084	8246979	210	Rock Chip	3.8	0.2	526
KJ 004	54	781993	8246723	205	Rock Chip	226	5.3	235
KJ 005	54	777701	8244591	149	Rock Chip	493	7.8	158
KJ 006	54	777963	8244646	150	Rock Chip	4.4	0.2	455
KJ 007	54	777963	8244646	151	Rock Chip	253	5.1	202
KJ 008	54	778410	8248727	164	Rock Chip	186	1.4	75

A reconnaissance survey targeting gold and base metals was also conducted during the same field campaign to ground-truth photogeological targets. Several samples were collected with analytical results still pending. Gold and base metals exploration will continue to be the Company's near term focus at the Cape Project.



Above: Buck white quartz vein to 2.5m width within greisen of the Chevy Creek Granite with pegmatitic textures on the contact margins.

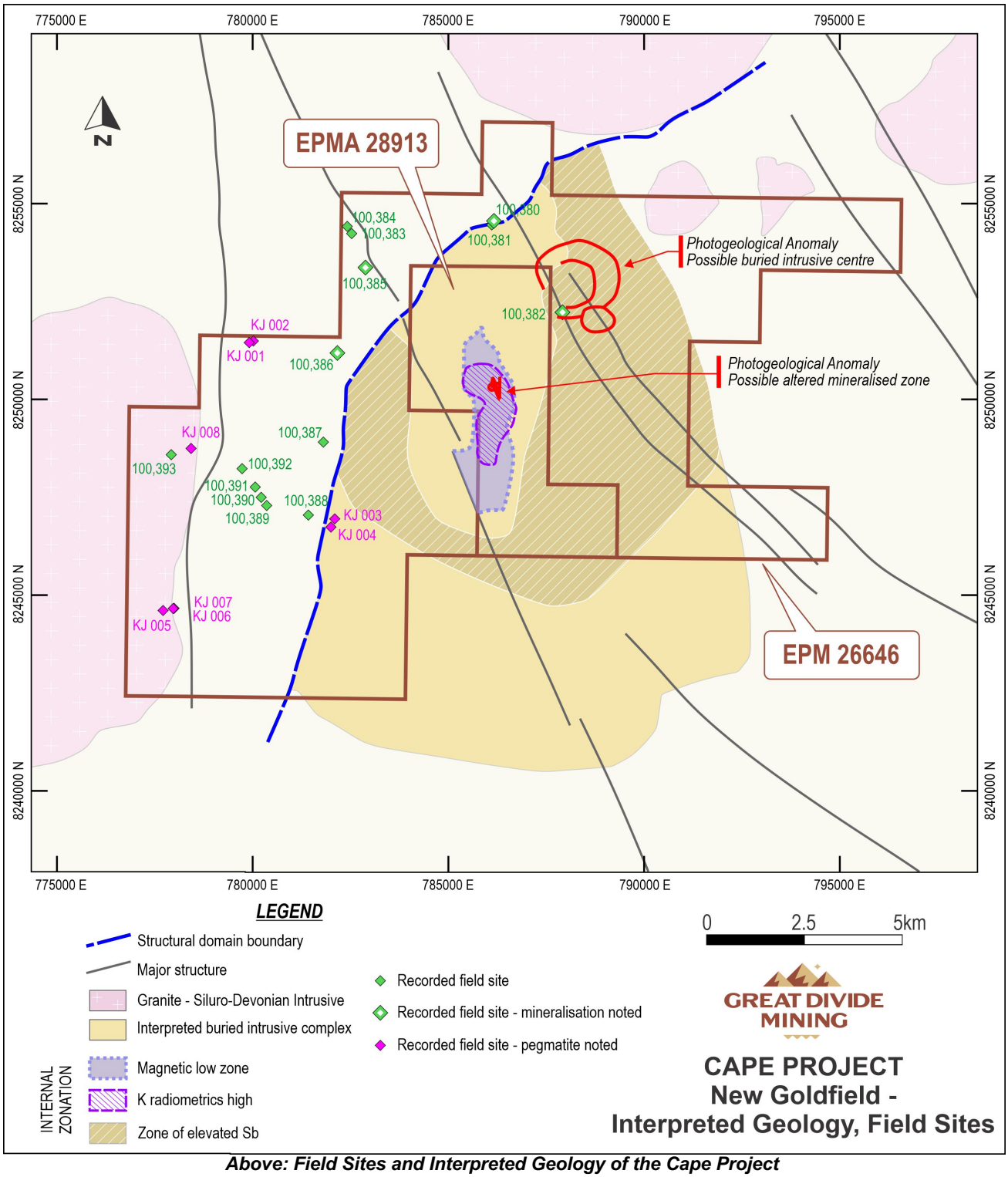
For personal use only

For personal use only



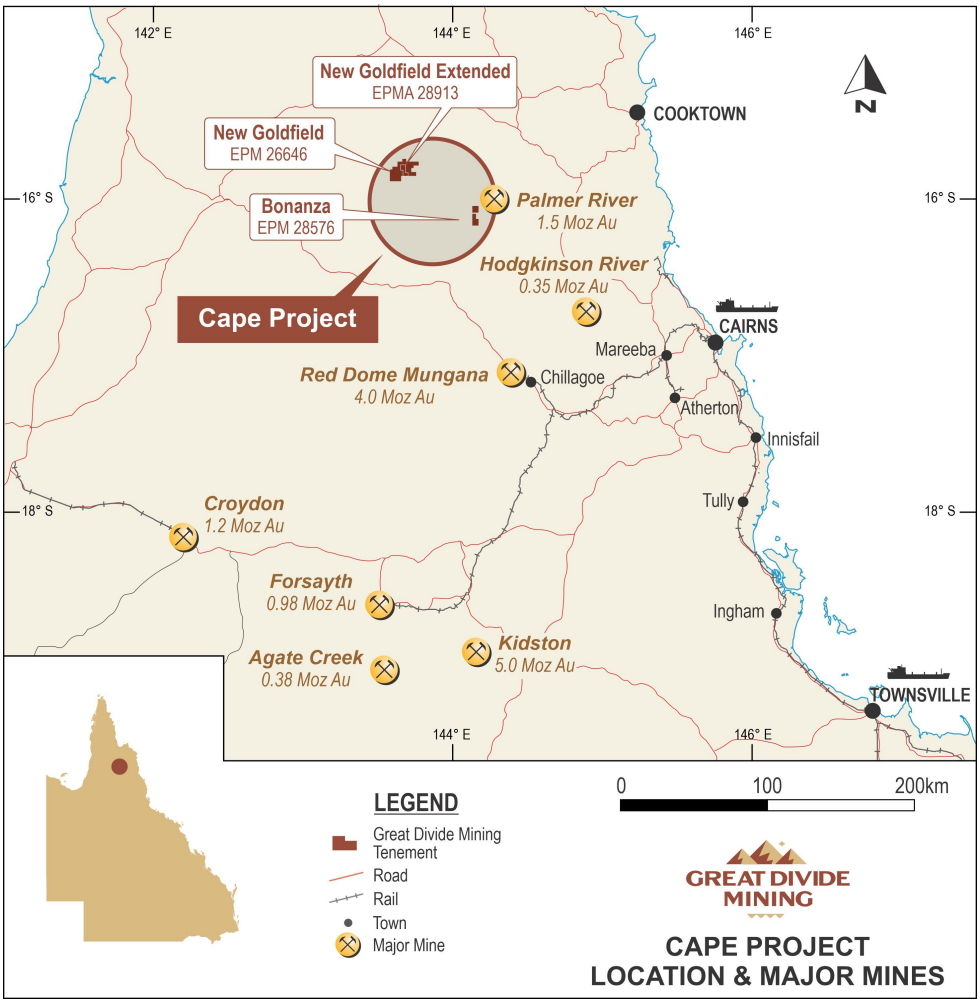
Above: Euhedral beryl and muscovite on contact margins of quartz vein within greisen of the Chevy Creek granite. The sample is approximately 7cm across.

For personal use only



Above: Field Sites and Interpreted Geology of the Cape Project

For personal use only



Above: Location map of the Cape Project, Queensland

GDM is also continuing with the exploration and assessment of its other projects.

The Yellow Jack geology and mining studies are continuing, assessing the commercial potential of the Project, with results expected in the coming months. Planning for exploration works at Coonambula and Devils Mountain is nearing completion in readiness for commencement of the field season.

A JORC Table 1 is annexed to this Announcement.

ENDS

ASX release authorised by the Board of Great Divide Mining Ltd.

For further information:

Justin Haines
Chief Executive Officer
e: justin.haines@greatdividemining.com.au

Investors and Media:

Reign Advisory
e: gdm@reignadvisory.com
p: +61 2 9174 5388

About Great Divide Mining Ltd (ASX: GDM)

Great Divide Mining is a Gold, Antimony and critical metals explorer in Queensland, with four projects across twelve tenements (including one in application). GDM's focus is on developing assets within areas of historical mining and past exploration with nearby infrastructure, thus enabling rapid development. Through a staged exploration and development programme, GDM intends to generate cash flow from its initial projects to support further exploration across its portfolio of highly prospective tenements.

Competent Persons Statement

The information in this announcement that relates to Exploration Results based on information compiled by Mr Justin Haines who is CEO of Great Divide Mining Ltd and a Fellow of the Australasian Institute of Mining and Metallurgy (AusIMM) and a Member of the Australian Institute of Geoscientists (AIG). Mr Haines has sufficient experience that is relevant to the styles of mineralisation and types of deposit under consideration, and to the activity that is being undertaking to qualify as a Competent Person, as defined in the 2012 Edition of the 'Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserves.' Mr Haines is an employee of GDM, and consents to the inclusion in this report of the matters based on his information in the form and context in which it appears.

Forward-Looking Statements

This announcement may contain forward-looking information about the Company and its operations. In certain cases, forward-looking information may be identified by such terms as "anticipates", "believes", "should", "could", "estimates", "target", "likely", "plan", "expects", "may", "intend", "shall", "will", or "would". These statements are based on information currently available to the Company and the Company provides no assurance that actual results will meet management's expectations. Forward-looking statements are subject to risk factors associated with the Company's business, many of which are beyond the control of the Company. It is believed that the expectations reflected in these statements are reasonable, but they may be affected by a variety of variables and changes in underlying assumptions which could cause actual results or trends to differ materially from those expressed or implied in such statements. There can be no assurance that actual outcomes will not differ materially from these statements.

For personal use only

For personal use only

JORC Table 1

Section 1 Sampling Techniques and Data

(Criteria in this section apply to all succeeding sections.)

Criteria	JORC Code explanation	Commentary
Sampling techniques	• *Nature and quality of sampling (eg cut channels, random chips, or specific specialised industry standard measurement tools appropriate to the minerals under investigation, such as down hole gamma sondes, or handheld XRF instruments, etc). These examples should not be taken as limiting the broad meaning of sampling.* • *Include reference to measures taken to ensure sample representivity and the appropriate calibration of any measurement tools or systems used.* • *Aspects of the determination of mineralisation that are Material to the Public Report.* • *In cases where 'industry standard' work has been done this would be relatively simple (eg 'reverse circulation drilling was used to obtain 1 m samples from which 3 kg was pulverised to produce a 30 g charge for fire assay'). In other cases more explanation may be required, such as where there is coarse gold that has inherent sampling problems. Unusual commodities or mineralisation types (eg submarine nodules) may warrant disclosure of detailed information.*	Great Divide Mining December 2023 program: • A total of 8 rock chip samples (KJ 001 to KJ 008) were collected by GDM's qualified geological consultants at the Cape Project in late 2023. • Samples are 1-3 kg rock chips taken from surface outcrops then placed in labelled calico sample bags. • All sample information, including lithological descriptions and GPS coordinates were recorded in the field during the sampling process. • Samples are taken from various locations and are not considered representative. • The accuracy of surface rock chip geochemistry is generally high but is generally not used in Mineral Resource estimations. Historical sampling: • Historical surface geochemical sampling includes surface rock chip, soil and stream sediment and pan concentrate samples from a number of different historical exploration companies working in the region between 1977 and 2009. • No information is available documenting measures to ensure sample representativity for surface sampling methods.
Drilling techniques	• *Drill type (eg core, reverse circulation, open-hole hammer, rotary air blast, auger, Bangka, sonic, etc) and details (eg core diameter, triple or standard tube, depth of diamond tails, face-sampling bit or other type, whether core is oriented and if so, by what method, etc).*	• No drilling programs are included.
Drill sample recovery	• *Method of recording and assessing core and chip sample recoveries and results assessed.* • *Measures taken to maximise sample recovery and*	• Not applicable

Great Divide Mining Ltd
ACN 655 868 803
GPO Box 154 Brisbane QLD 4001
greatdividemining.com.au

For personal use only

Criteria	JORC Code explanation	Commentary
	ensure representative nature of the samples. • *Whether a relationship exists between sample recovery and grade and whether sample bias may have occurred due to preferential loss/gain of fine/coarse material.*	
Logging	• *Whether core and chip samples have been geologically and geotechnically logged to a level of detail to support appropriate Mineral Resource estimation, mining studies and metallurgical studies.* • *Whether logging is qualitative or quantitative in nature. Core (or costean, channel, etc) photography.* • *The total length and percentage of the relevant intersections logged.*	Great Divide Mining December 2023 program: • Rock chip samples were geologically logged in the field by an experienced geologist. • Rock chip samples were single spot outcrop samples, taken from various locations (see Map and Table in body of report). • Descriptions of samples are mostly qualitative (e.g. lithology, alteration, veining and mineralisation) with limited quantitative logging of key minerals or veins. Historical sampling: • Descriptions of geochemical samples (where available) are mostly qualitative (e.g. lithology, alteration, veining and mineralisation) with no quantitative logging of key minerals or veins.
Sub-sampling techniques and sample preparation	• *If core, whether cut or sawn and whether quarter, half or all core taken.* • *If non-core, whether riffled, tube sampled, rotary split, etc and whether sampled wet or dry.* • *For all sample types, the nature, quality and appropriateness of the sample preparation technique.* • *Quality control procedures adopted for all sub-sampling stages to maximise representivity of samples.* • *Measures taken to ensure that the sampling is representative of the in situ material collected, including for instance results for field duplicate/second-half sampling.* • *Whether sample sizes are appropriate to the grain size of the material being sampled.*	Great Divide Mining December 2023 program: • The 8 rock chip samples (1-3 kg) were collected in the field from outcrops using a geological hammer. • Samples were despatched to SGS Laboratory, a certified commercial laboratory, who carried out appropriate sample preparation methods. • Samples were dried, crushed and pulverised, as per standard industry practice. Historical sampling: • It has been assumed that sample preparation methods used by all commercial laboratories followed the basic steps of drying, crushing, and pulverising, but details of the amount of the sample crushed and pulverised are not known. Therefore, it is not possible to assess the quality and appropriateness of the sample preparation techniques. No information is available on the size of the samples submitted for analysis. • No information has been recorded that documents quality control procedures adopted for all sub-sampling stages to maximise representivity

For personal use only

Criteria	JORC Code explanation	Commentary
		of samples. • No information has been recorded that documents measures taken to ensure that the sampling is representative of the in-situ material collected. • No formal assessment has been undertaken to quantify the appropriate sample size required for good quality determination of gold content, given the nature of the gold mineralisation.
Quality of assay data and laboratory tests	• *The nature, quality and appropriateness of the assaying and laboratory procedures used and whether the technique is considered partial or total.* • *For geophysical tools, spectrometers, handheld XRF instruments, etc, the parameters used in determining the analysis including instrument make and model, reading times, calibrations factors applied and their derivation, etc.* • *Nature of quality control procedures adopted (eg standards, blanks, duplicates, external laboratory checks) and whether acceptable levels of accuracy (ie lack of bias) and precision have been established.*	Great Divide Mining December 2023 program: • All 8 samples were analysed at the commercial laboratory SGS using standard analytical techniques. • Samples were assayed for gold using the SGS GO_FAA30V10 technique. Samples were also assayed for a multi-element suite using the GE_IMS92A50 and GE_ICP92A50 analytical techniques. • No geophysical tools, spectrometers, or handheld XRF instruments have been used to date. • Due to the small batch and early-stage nature of the sampling program, no QAQC (standard, duplicate or blank) samples were inserted into the sample batch. • The lack of quality control procedures is considered acceptable for this early-stage exploration where the results are not being used for Mineral Resource estimation. Historical sampling: • Baron rock chips: Samples were analysed for gold but the laboratory details and assay methods are unknown. • CRA stream sediment and rock chip: Samples were analysed at Classic Laboratories, Townsville for gold by fire assay. Rock chip samples were also analysed for Ag, Co, Cd, Cr, Cu, Fe, Ni, P, Sb, Sn, Pb, V and Zn by AAS finish and As, Ba, Mo, Pb and W by XRF. • Wyala Resources NL stream sediment: Stream sediment pan concentrate samples were analysed at Classic Comlabs, Townsville. Gold and Fe was analysed by aqua regia digest and AAS determination while REE (Ce, La, Y) were analysed by XRF. • No geophysical tools, spectrometers, or handheld XRF instruments were used. • Limited details of the use of QAQC (duplicates, blanks, standards or

For personal use only

Criteria	JORC Code explanation	Commentary
		certified reference materials) have been reported.
Verification of sampling and assaying	• *The verification of significant intersections by either independent or alternative company personnel.* • *The use of twinned holes.* • *Documentation of primary data, data entry procedures, data verification, data storage (physical and electronic) protocols.* • *Discuss any adjustment to assay data.*	Great Divide Mining December 2023 program: • GDM has collated and created a digital database of all exploration completed at the Project. • No adjustments to assay data have been made. • GDM is awaiting further assay results. Historical sampling: • It has not been possible to independently verify any significant grades reported. • No adjustments to assay data have been made.
Location of data points	• *Accuracy and quality of surveys used to locate drill holes (collar and down-hole surveys), trenches, mine workings and other locations used in Mineral Resource estimation.* • *Specification of the grid system used.* • *Quality and adequacy of topographic control.*	Great Divide Mining December 2023 program: • Sample sites were surveyed using hand-held GPS, with a +/- 3m to 5m accuracy. • The coordinate system used is Geocentric Datum of Australia (GDA2020) in Map Grid of Australia (MGA) zone 54. • Quality of the topographic control data is reliant on public domain topographic data. • A sample location map and sample details table are included in the body of the report, Historical sampling: • No details of the accuracy and quality of surveys used to locate samples is recorded. • Various coordinate systems were used by previous explorers, then converted by GDM to Geocentric Datum of Australia (GDA2020) in Map Grid of Australia (MGA) zone 54. • Quality of the topographic control data is reliant on public domain data.
Data spacing and distribution	• *Data spacing for reporting of Exploration Results.* • *Whether the data spacing and distribution is sufficient to establish the degree of geological and grade continuity appropriate for the Mineral Resource and Ore Reserve estimation procedure(s) and classifications applied.*	Great Divide Mining December 2023 program: • The spacing of data is variable and controlled by the limited availability of outcrop. • There are no reported Mineral Resources or Reserves - the sample results will not be used for Mineral Resource and Ore Reserve estimation.

For personal use only

Criteria	JORC Code explanation	Commentary
	• *Whether sample compositing has been applied.*	• No sample compositing was carried out on site. Historical sampling: • The spacing of data is variable. • There were no reported Mineral Resources or Reserves. • The sample results will not be used for Mineral Resource and Ore Reserve estimation. • No information is available documenting sample compositing.
Orientation of data in relation to geological structure	• *Whether the orientation of sampling achieves unbiased sampling of possible structures and the extent to which this is known, considering the deposit type.* • *If the relationship between the drilling orientation and the orientation of key mineralised structures is considered to have introduced a sampling bias, this should be assessed and reported if material.*	• Surface sampling techniques are considered appropriate for the early-stage exploration. • Drilling will be required to establish the optimal orientation. • No sampling bias is considered to have been introduced in sampling completed to date.
Sample security	• *The measures taken to ensure sample security.*	Great Divide Mining December 2023 program: • At the completion of the field program, the samples were transported by GDM's Consultant to the Consultant's office, then by courier to SGS laboratories. Historical sampling: • No chain of custody is documented for previous sampling.
Audits or reviews	• *The results of any audits or reviews of sampling techniques and data.*	Great Divide Mining December 2023 program: • The Consultant's report on the exploration results were audited internally by GDM. Historical sampling: • Derisk Geomining Consultants completed a review of the historical exploration results from this project in 2023.

For personal use only

Section 2 Reporting of Exploration Results

(Criteria listed in the preceding section also apply to this section.)

Criteria	JORC Code explanation	Commentary
Mineral tenement and land tenure status	• *Type, reference name/number, location and ownership including agreements or material issues with third parties such as joint ventures, partnerships, overriding royalties, native title interests, historical sites, wilderness or national park and environmental settings.* • *The security of the tenure held at the time of reporting along with any known impediments to obtaining a licence to operate in the area.*	• The Project tenement comprises EPM 26646 and EPM Application 28913. These tenements are currently held 100% by Muscovite Gold Exploration Pty Ltd, a 100% subsidiary of Great Divide Mining Ltd. • Refer to the tenement table in the Company's Annual Report dated November 2023. • The tenements are in good standing.
Exploration done by other parties	• *Acknowledgment and appraisal of exploration by other parties.*	• Numerous historical exploration permits have been held over parts and/or all the Project area. Previous exploration has included geological mapping, stream sediment, soil and rock chip geochemical sampling and airborne geophysics. The main programs included: ○ Comalco Ltd (1977 - 1978) completed geological mapping and geochemical surveys. ○ Geopeko Ltd (1979 – 1980) completed geological mapping, geochemical surveys, and airborne geophysics. ○ Baron Gold Ltd (1981 – 1985) completed geological mapping and geochemical surveys. ○ Wyala Resources NL (1987 – 1990) completed geological mapping and geochemical surveys. ○ CRA Exploration Ltd (1991 – 1993) completed geochemical surveys. ○ Mt Isa Mines Ltd (1991 – 1993) completed geological mapping, geochemical surveys and evaluation of magnetic anomalies. ○ BHP Minerals Ltd (1996 – 1997) completed geological mapping, geochemical surveys and airborne geophysics. ○ BHP Billiton Ltd (2005 – 2006) completed an airborne EM survey. ○ Energy Minerals Pty Ltd (2006 – 2009) completed geological mapping and geochemical surveys. ○ Delminco Pty Ltd (2007 – 2009) completed literature reviews.
Geology	• *Deposit type, geological setting and style of mineralisation.*	• The New Goldfield Project (within the Cape Project) is located within the Yambo Subprovince of the Etheridge Province which crops out over a significant proportion of north Queensland, extending from Woolgar in the south to Lockhart River in the north. • GDM consider that the Project is prospective for:

For personal use only

Criteria	JORC Code explanation	Commentary
		○ Intrusion-related gold deposits and mesothermal quartz vein or orogenic gold mineralisation). The district contains numerous old gold mine workings and known mineral occurrences. ○ Pegmatite or intrusive hosted lithium and rare earth element (REE) mineralisation.
Drill hole Information	• *A summary of all information material to the understanding of the exploration results including a tabulation of the following information for all Material drill holes:* ○ *easting and northing of the drill hole collar* ○ *elevation or RL (Reduced Level – elevation above sea level in metres) of the drill hole collar* ○ *dip and azimuth of the hole* ○ *down hole length and interception depth* ○ *hole length.* • *If the exclusion of this information is justified on the basis that the information is not Material and this exclusion does not detract from the understanding of the report, the Competent Person should clearly explain why this is the case.*	• Not applicable.
Data aggregation methods	• *In reporting Exploration Results, weighting averaging techniques, maximum and/or minimum grade truncations (eg cutting of high grades) and cut-off grades are usually Material and should be stated.* • *Where aggregate intercepts incorporate short lengths of high grade results and longer lengths of low grade results, the procedure used for such aggregation should be stated and some typical examples of such aggregations should be shown in detail.* • *The assumptions used for any reporting of metal equivalent values should be clearly stated.*	• No capping of high grades was performed. • No details of the aggregation of data was recorded for historical exploration results. • No metal equivalents are reported.
Relationship between mineralisation widths and intercept lengths	• *These relationships are particularly important in the reporting of Exploration Results.* • *If the geometry of the mineralisation with respect to the drill hole angle is known, its nature should be reported.* • *If it is not known and only the down hole lengths are reported, there should be a clear statement to*	• Rock chip samples were taken from surface outcrop and are not representative of the entire thickness of the underlying rock units.

For personal use only

Criteria	JORC Code explanation	Commentary
	this effect (eg 'down hole length, true width not known').	
Diagrams	• *Appropriate maps and sections (with scales) and tabulations of intercepts should be included for any significant discovery being reported These should include, but not be limited to a plan view of drill hole collar locations and appropriate sectional views.*	• Plans of sample locations have been provided.
Balanced reporting	• *Where comprehensive reporting of all Exploration Results is not practicable, representative reporting of both low and high grades and/or widths should be practiced to avoid misleading reporting of Exploration Results.*	• Balanced reporting of Exploration Results is presented.
Other substantive exploration data	• *Other exploration data, if meaningful and material, should be reported including (but not limited to): geological observations; geophysical survey results; geochemical survey results; bulk samples – size and method of treatment; metallurgical test results; bulk density, groundwater, geotechnical and rock characteristics; potential deleterious or contaminating substances.*	• The Project includes a moderate amount of exploration data collected by previous companies, including regional stream sediment geochemical data, pan concentrates, soil sample and rock chip data, geological mapping data and geophysical survey data. Much of this data has been captured and validated into a GIS database. • There is no other exploration data that is considered to be material to the results reported herein.
Further work	• *The nature and scale of planned further work (eg tests for lateral extensions or depth extensions or large-scale step-out drilling).* • *Diagrams clearly highlighting the areas of possible extensions, including the main geological interpretations and future drilling areas, provided this information is not commercially sensitive.*	• Great Divide Mining plans to conduct further surface geological mapping, geochemical sampling, ground geophysics and drilling across various high-priority target areas over the next two years.